

PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 1 · How this class works, and a statistics review

TUESDAY AUG 25 AND THURSDAY AUG 27, 2026

INTRODUCE YOURSELF.

Name, major, and year.

Why you registered for this class.

The sport you follow most closely, and the job you would take tomorrow if somebody offered it.

01

HOW THIS CLASS WORKS

MY GOAL: EQUIP YOU WITH THE INTUITION, CRITICAL THINKING, TECHNICAL, AND COMMUNICATION SKILLS NEEDED TO USE ANALYTICS TO ADD VALUE TO AN ORGANIZATION.

EVERY UNIT HAS FOUR PARTS

1 Tuesday's lecture.

The unit's concepts, in class.

2 Thursday, in class, by hand, no agentic AI.

We will use Excel to do the calculations and build an understanding of our approaches. Without a mental model of what we're doing and why, you will misapply models to the wrong contexts and communicate their outputs incorrectly. You may still use an LLM to assist with troubleshooting formulas and understanding our approach.

3 The take-home Case Study. On your own, with Claude Code.

You will practice using agentic coding tools to quickly run similar analysis more efficiently. When complete, you will communicate your findings in your own words, along with submitting the code.

4 Tuesday's quiz, closed-book.

A short quiz at the start of Tuesday's class to confirm that you understand the concepts.

ASSIGNMENT ASSISTANCE RESTRICTIONS

1 Thursday, in class, by hand, no agentic AI.

You may use an LLM to assist with troubleshooting formulas and understanding our approach. The hand-build may be finished at home that evening; it is submitted by Monday at 11:59pm, along with the AI-assisted assignment and brief.

2 The take-home Case Study. On your own, with Claude Code.

Not allowed: submitting analysis created or edited by another human. You may discuss assignments with classmates, and use books, instructors, and LLMs. Not allowed: submitting briefs written by AI or another human. The BRIEF.md submissions must be in your own words; copying and pasting from an LLM and changing some of the verbiage is also in violation. You need to be able to sit in a room with your boss, a coach, or a GM and explain your findings.

3 Tuesday's quiz, closed-book.

Quizzes and both exams are closed-book and proctored. If you can only explain a concept by consulting an outside resource, you don't understand the concept yet.

GRADING

COMPONENT	WEIGHT	NOTES
Quizzes	25%	Weekly, Tuesday, closed-book. Two lowest dropped.
Case studies	25%	Equal halves: the analysis and the brief. Audit and quiz artifacts required.
Exams	25%	Exam 1 written (Oct 8). Exam 2 is a live audit (Nov 19).
Final project	25%	Proposal Week 11, presented Dec 8. No final exam.

FOUR COMPONENTS OF EACH CASE STUDY

1 The analysis.

The spreadsheet, code, chart, table. Whatever you built, including Thursday's hand-built file.

2 The quiz transcript.

`/quiz-me` quizzes you on that week's concepts and writes the whole exchange into your repo. This will prepare you for the in-class quiz on Tuesday.

3 The audit record.

`/audit` reviews your analysis like a skeptical senior analyst and writes down each finding. You respond to every one – fix it, or defend why it stands – in the same file.

4 The brief.

`BRIEF.md`, half a page to the week's audience, typed by you.

HOW A CASE STUDY IS GRADED

1 Half: the analysis.

The spreadsheet, code, chart, table. Whatever you built, including Thursday's committed hand-built file.

2 Half: BRIEF.md, typed by you.

Half a page to the week's named audience, usually a coach. Graded on decision quality and on honesty about what you do not know. You may ask Claude to red-team the draft; you may not ask it to write the draft.

3 Required but not graded: the audit record and quiz transcript.

These steps will help catch an issue with your analysis before I do. They'll also prepare you for your quiz.

THE FIRST HALF OF THE SEMESTER

WK	TUE / THU	CONTENT
1	Aug 25 / 27	Onboarding and statistics review
2	Sep 1 / 3	Classical probability and inference
3	Sep 8 / 10	Bayesian statistics
4	Sep 15 / 17	Descriptive and probabilistic statistics
5	Sep 22 / 24	Introduction to regression
6	Sep 29 / Oct 1	Regression continued
7	Oct 6 / 8	Review Tuesday. Exam 1 Thursday.

First case study due Mon Aug 31. Labor Day pushes Week 2's to Tue Sep 8. Sep 8 is the last add or drop day.

THE SECOND HALF OF THE SEMESTER

WK	TUE / THU	CONTENT
8	Oct 13 / 15	Data collection and responsible automation
9	Oct 20 / 22	Storytelling, visualization, skills and hooks
10	Oct 27 / 29	Logical fallacies and AI shortcomings
11	Thu Nov 5 only	Monte Carlo. Final project proposal due.
12	Nov 10 / 12	Machine learning
13	Nov 17 / 19	Practical exam: three flawed analyses, audited live
14	Dec 1 / 3	Final project work sessions
15	Tue Dec 8	Final presentations

Election Day suspends classes Tue Nov 3. No class Nov 23 to 27 for Thanksgiving. Final deliverable due Mon Dec 14.

02

OUR TECH STACK

Today is setup. Thursday you use all of it on real data.

THREE THINGS BEFORE YOU LEAVE TODAY

1 Claude Pro, installed and signed in.

Roughly twenty dollars a month for four months. If that is a real obstacle, email me privately and it gets handled. Nobody is dropping this course over a subscription.

2 Claude Code and Git on your machine.

Half an hour with the setup page, longer if you have never used a terminal. Do it before class if you can. Stragglers get fixed Thursday, not left behind.

3 Your repository, created from the course template on GitHub.

This is where every assignment lives for the rest of the semester. Your copy is yours; I can see it, your classmates cannot.

THE SAME FIVE MOVES EVERY WEEK

1 Hand-build the method.

Thursday, in class, no agentic AI. One team, ten trials, two hypotheses.

2 Scale it with the agent.

Your artifact is the specification: the target, the assumptions, the transformations, the evaluation, the outputs you want. Claude writes the Python. You read the outputs, not the code.

3 Verify adversarially.

Run `/audit`. Check the agent's answer against your Thursday hand-build. If they disagree, one of you is wrong.

4 Brief the week's audience.

BRIEF.md, half a page, typed by you. What you found, what you would do, and what would change your mind.

5 When you are done, run `/submit`.

It checks the whole week – hand-build, analysis, audit, brief, quiz – tells you what is missing, and pushes when it is all there. Repo link in Canvas by Monday 11:59pm.

THE FOUR COMMANDS YOU WILL USE MOST

/teach

Explains a concept using the week's data.

/audit

Has an agent audit your own work. Uses a subagent to prevent reasoning context from biasing it towards its own approach. Run this before you push your assignments.

/quiz-me

Quizzes you on that week's concepts before you submit your assignment. This will prepare you for the in-class quiz on Tuesday.

/submit

Submits your work. Before doing so, it checks every required file and tells you if anything's missing.

SETUP TIME.

- 1 Install Git and Python.
- 2 Install the Claude Desktop app. Sign in.
- 3 GitHub account. Create your repo from the template. Add jackdav1.
- 4 Install the Python packages.
- 5 First commit, pushed.

Full instructions: SPAX-402-Setup on Canvas. Flag me when something breaks – we continue once everyone is running.

03

THE SHAPE OF THE DATA

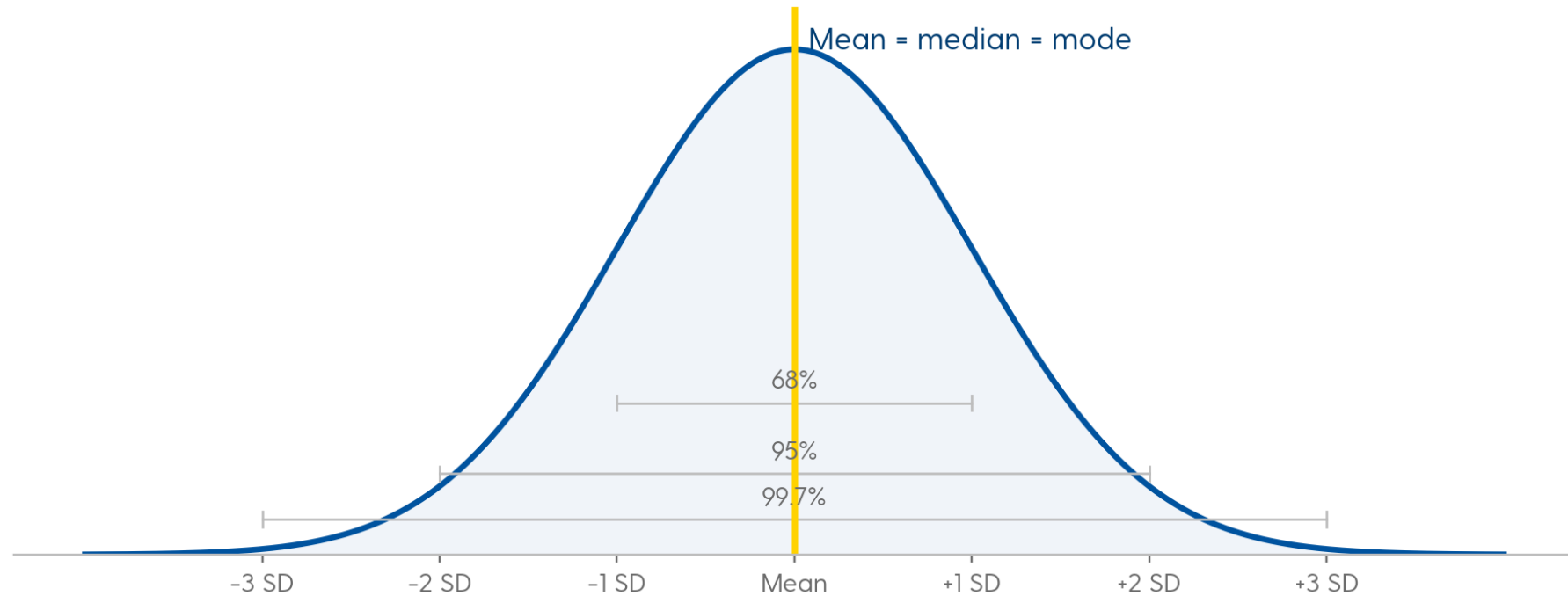
Mean, median, mode, and the reason the first question about any statistic is what its distribution looks like.

MEAN, MEDIAN, MODE

The average, the middle value, and the most common value.
In a symmetric distribution they land on the same number;
how far apart they sit is a measure of how skewed the
data is.

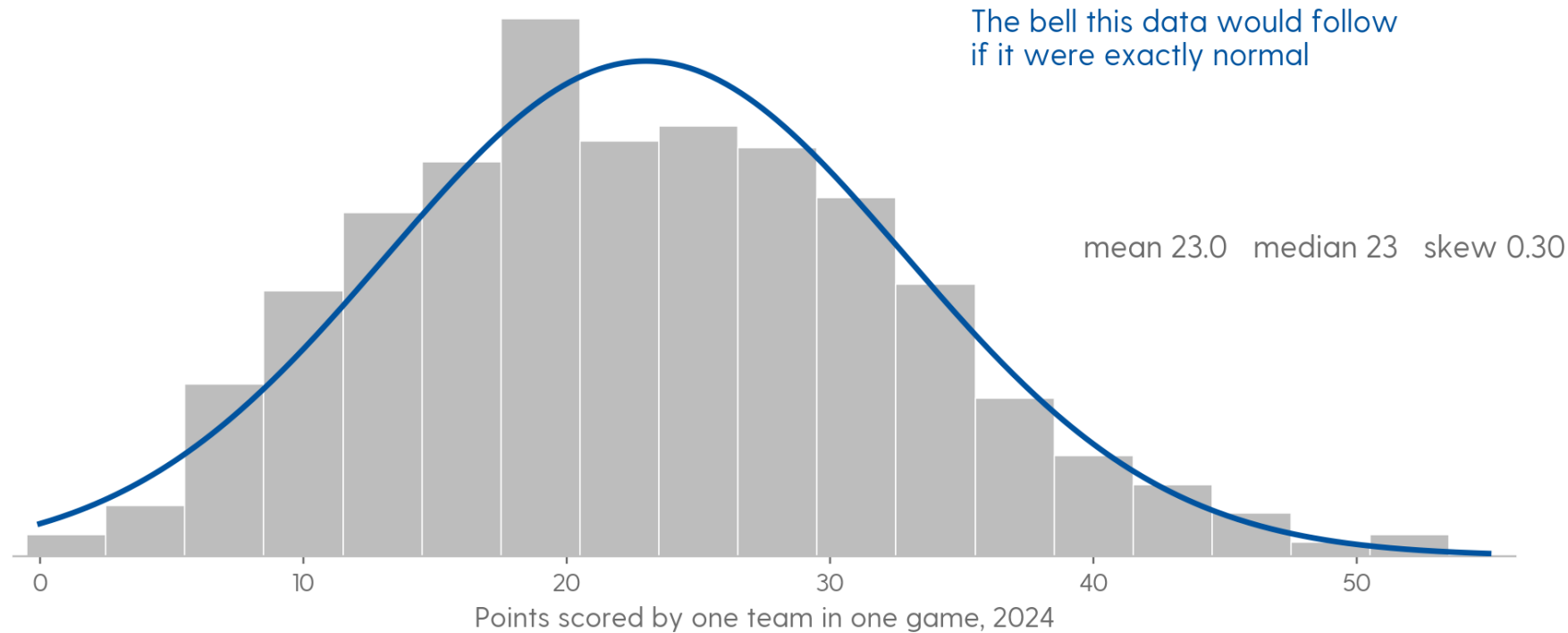
Mean = $\sum x / n$. Median = the 50th percentile. Mode = the most frequent value, and the only one of the three that also works on categories.

WHERE THE THREE-SIGMA RULES COME FROM



Symmetric, one peak, mean equals median equals mode. Roughly 68, 95, and 99.7 percent of values inside one, two, and three standard deviations. Every one of those numbers is a property of this exact shape.

POINTS PER TEAM-GAME IS CLOSE TO NORMAL



Mean 23.01, median 23, skew 0.30 across 570 team-games. Close enough to normal that the sigma rules roughly hold. Most sport statistics are not.

nflverse play-by-play, 2024 regular season and playoffs

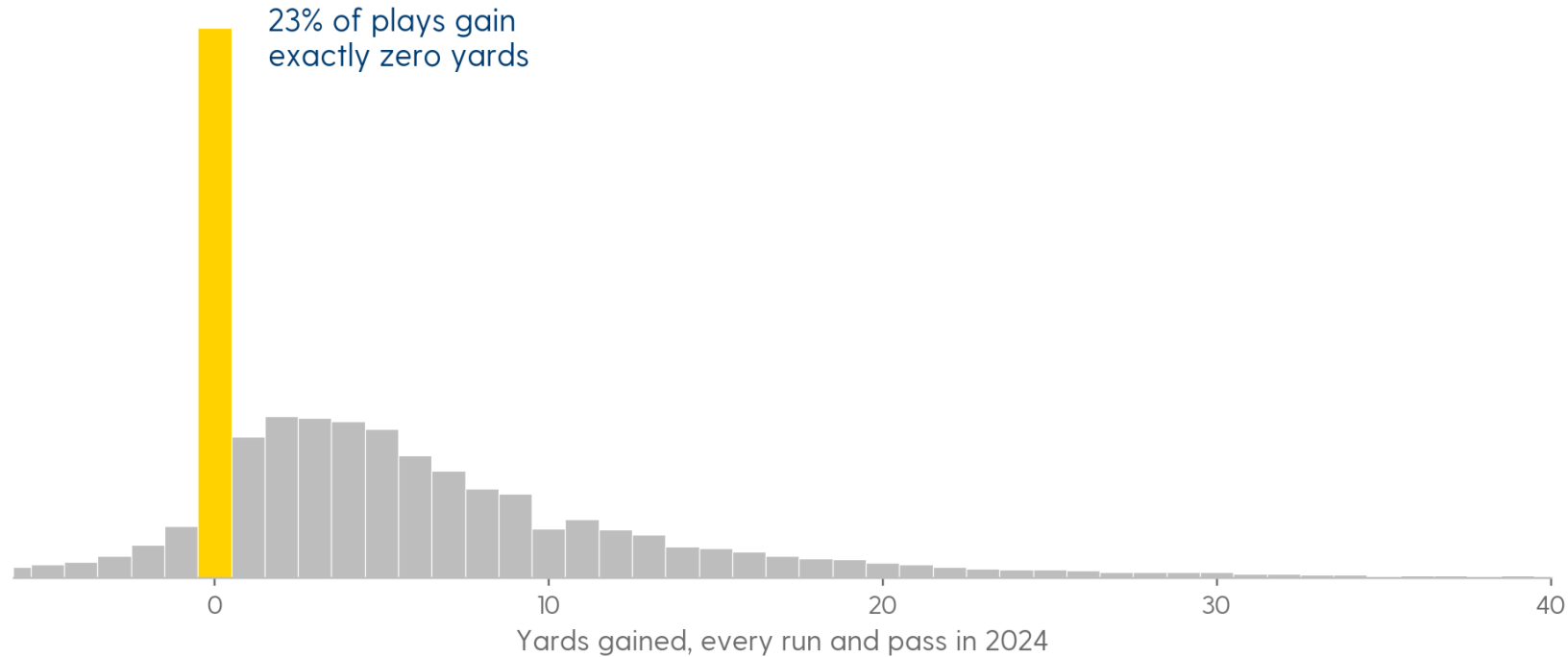
? RUSHING YARDS PER CARRY: DO YOU EXPECT THE MEAN TO BE HIGHER OR LOWER THAN THE MEDIAN?

WHICH OF THESE STATISTICS DO YOU EXPECT TO BE SKEWED, IN WHICH DIRECTION, AND WHY?

Home runs. Free-throw percentage. Golf scores. Time to recover from an injury.
Height. Salary.

For each one: is there a floor, is there a ceiling, and what would it take for a single athlete to sit far out on the tail?

TWO PEAKS USUALLY MEAN YOU HAVE MIXED TWO DIFFERENT THINGS TOGETHER



22.8% of all runs and passes gain exactly zero yards. That spike is a second population hiding inside the first.

nflverse play-by-play, 2024 regular season and playoffs

 **WHAT DO YOU THINK THE SPIKE AT EXACTLY ZERO YARDS IS?**

A BIMODAL DISTRIBUTION MEANS SOMETHING IN YOUR DATA IS TWO THINGS.

Find the subclass before you compute anything else. Every summary statistic you calculate before you split is an average of two populations that do not belong in the same average.

04

SPREAD, AND WHY THE RAW NUMBER LIES

Standard deviation, the coefficient of variation, and a ranking that reverses depending on which one you use.

STANDARD DEVIATION

Roughly the average distance of a data point from the mean. Small means the values cluster; large means they scatter.

The square root of the mean squared distance from the mean. Squaring is what stops the positive and negative distances from cancelling, and it is also why one extreme value moves an SD more than it moves a median.

SAY IT TO A COACH

How would you put this in one sentence?

COEFFICIENT OF VARIATION

The standard deviation divided by the mean, so spread can be compared across statistics measured on completely different scales.

CV = SD / mean . Unitless, so it survives a change of units. Meaningless when the mean sits near zero, and misleading on anything that can go negative.

SAY IT TO A COACH

How would you put this in one sentence?

 **WHAT LEAGUE HAS THE MOST VARIATION IN SCORING?**

05

SPREAD WITH NO NUMBERS TO AVERAGE

Standard deviation needs quantities. Play calls, pitch types, and formations are categories, and they need a different measure.

YOU CANNOT TAKE THE STANDARD DEVIATION OF “RUN, PASS, PASS, RUN.”

There is nothing to average. But a play-caller who runs on every short-yardage snap is obviously more predictable than one who splits it fifty-fifty, so the variation is real and it needs a number.

ENTROPY

ENTROPY MEASURES HOW SPREAD OUT A SET OF SHARES IS

$H = - \sum p \cdot \ln(p)$ over every category

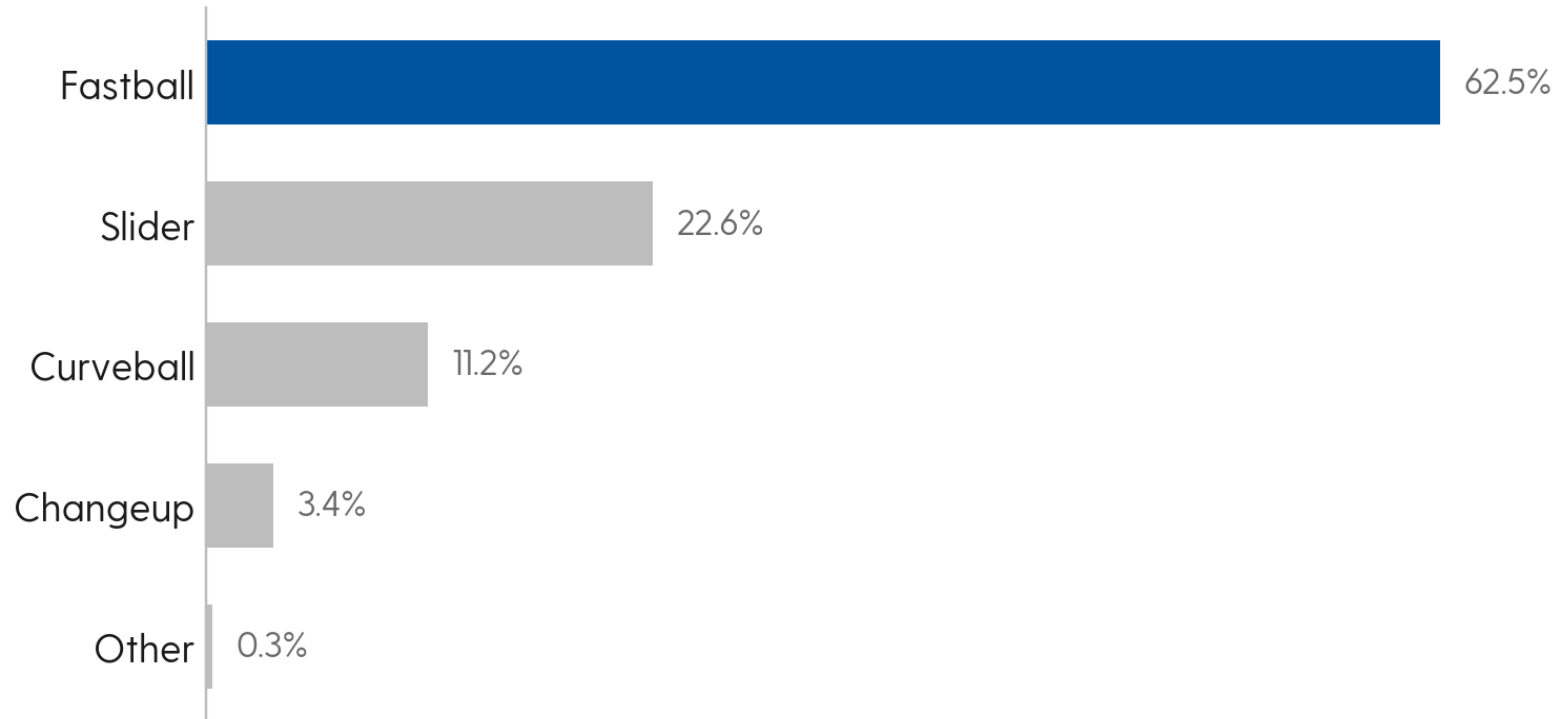
Max $H = \ln(n)$ the value when every share is equal

Standardized entropy = $H / \ln(n)$

A category with a share of zero contributes zero, not an error. Do not try to take the log of nothing.

RANGES 0 TO 1: 0 IS FULLY PREDICTABLE, 1 IS AN EVEN SPLIT.

KERSHAW THREW ONE PITCH 62.5% OF THE TIME



Two more pitches, the sinker and the cutter, are in the arsenal and were thrown zero times in this sample. Whether they count toward n is a decision you make and then state.

Published Statcast pitch-mix shares

? ON THE ZERO-TO-ONE SCALE, WHERE DOES THIS PITCH MIX LAND?

06

THIS WEEK'S WORK

Two case studies, the same four moves, and a result that does not come out the way you would expect.

CASE STUDY: ENTROPY IN THE NFL

1 Thursday, by hand: one team.

Standardized entropy for a single team's run/pass mix on third and fourth and short, from the formula, on paper. Keep it.

2 Take-home, with the agent: all thirty-two.

Pull the play-by-play with the provided script, and have the agent explain to you how that script limits and caches its requests. Filter, compute each team's split, entropy, and conversion rate.

3 Verify before you accept anything.

The agent has to reproduce your Thursday team and match your number. If it cannot, do not accept the league table.

4 Ship the table and one chart.

Team table to outputs/ as .xlsx, plus a chart relating entropy to conversion rate. Then /audit, then /quiz-me, then /submit.



**COULD A TEAM'S ENTROPY COME OUT AT EXACTLY ZERO? WHAT
WOULD THAT TAKE?**

? DO THE LEAST PREDICTABLE TEAMS CONVERT SHORT YARDAGE MORE OFTEN?

CASE STUDY: NBA FIELD-GOAL PERCENTAGE BY POSITION

1 Same four moves, second dataset.

Thursday's hand-build covers one position. The agent version covers every position, and the hand-build is how you check it.

2 Describe the shape before you compare the groups.

Is it unimodal or bimodal? Skewed which way? If it is bimodal, what are the two populations, and does comparing positions even make sense before you split them?

3 Highest FG% relative to position is a different question from highest FG%.

Which one a coach wants depends on the decision, and your brief should say which decision you assumed.

WHAT GOES IN THE BRIEF.

Do you see a relationship between unpredictable play-calling and converting short-yardage downs? Which team is the interesting outlier, and what is your best explanation?

What else would you want to know before telling a team to change its run/pass mix?

Beyond raw FG%, how would you compare scoring efficiency across positions?

Half a page, in BRIEF.md, due Monday at 11:59pm.

THE TAKE-HOME, STEP BY STEP

- 1** **The question.**
NFL: do less predictable play-callers convert short yardage more often? NBA: what shape is the league's FG% distribution? What subclass might be causing this shape? How should it factor into how we compare players?
- 2** **The data.**
NFL: nflverse play-by-play, 2024 season. The pull script is already in your repo at weeks/week01/data/. NBA: league FG% by player and position.
- 3** **Open Claude Code in your repo.**
Spec the job in your own words: the question, the target, the assumptions. (/teach is optional, if you want the week's concepts walked through first.)
- 4** **Direct the agent.**
Run the pull script, filter to third and fourth and short, compute each team's run/pass split, standardized entropy, and conversion rate.
- 5** **Verify against Thursday.**
Put your hand-build into weeks/week01/ in your repo: the Excel file, or a photo of the paper work. The agent must reproduce your team and match your number before you accept the league table.
- 6** **Ship it.**
Ask Claude to save your team table as .xlsx in outputs/, plus one chart. Write BRIEF.md in your own words. Run /audit and /quiz-me, then /submit, which checks nothing is missing and pushes to GitHub. Finally, add your repo link to Canvas by Monday 11:59pm.

BEFORE THURSDAY

Claude Pro, Claude Code, and Git installed. Your repo created from the template, with me added as collaborator, and MISSION.md filled in. That is your first commit.

Read the Week 1 quiz bank in your repo. It is not a secret, and the quiz comes out of it.

Bring your laptop with Excel, and paper for the entropy calculation. Finish the hand-build early and you start the case study in class instead of at home.